**Focused ultrasound–mediated AAV gene therapy: Unlocking a new space in the CNS delivery landscape**

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## Abstract

Adeno-associated viruses (AAVs) have emerged as the most clinically advanced platform for *in vivo* gene delivery, enabling durable therapeutic gene expression across a growing number of neurological disorders. However, successful delivery of AAVs to the brain requires navigating trade-offs among three interdependent factors: the vector dose administered, the resulting delivery efficacy within target cells, and the volume of brain tissue that can be effectively covered. While multiple clinical strategies have demonstrated that very different positions within this delivery landscape can achieve therapeutic benefit, each approach carries important limitations, including surgical invasiveness, limited regional coverage, peripheral toxicity at high doses, manufacturing burden, or the risks associated with global brain transgene exposure.

This review situates focused ultrasound–mediated blood-brain barrier opening (FUS-BBBO) within this broader CNS delivery landscape. Through the distinct mechanical mechanism of microbubble cavitation, FUS transiently and locally increases BBB permeability at user-defined targets, enabling systemically administered AAVs to enter the brain at high local concentrations while sparing the remainder of the brain and vasculature. We discuss how advances in large-volume sonication and FUS-optimized AAV capsids are converging to enable a clinically translatable delivery paradigm that is non-invasive, regionally selective, and highly efficient.

## Introduction

Adeno-associated viruses (AAVs) have emerged as the most clinically advanced vector platform for *in vivo* gene delivery, enabling long-term expression of therapeutic genes across a broad range of diseases[1–3]. Over the past two decades, dozens of clinical trials have explored AAV-mediated therapies for neurological, metabolic, neuromuscular, and ophthalmic conditions, reflecting both the versatility of AAV biology and the growing clinical need for durable genetic treatments. While there are currently seven FDA approved AAV-based drugs, only two of these are approved for the CNS: onasemnogene abeparvovec (Zolgensma) for spinal muscular atrophy[4] and eladocagene exuparvovec (Kebilidi) for aromatic L-amino acid decarboxylase (AADC) deficiency[5]. These two therapies occupy opposite ends of the delivery spectrum—Zolgensma is administered systemically at an exceptionally high dose to treat a diffuse neuromuscular disorder, whereas Kebilidi is delivered through direct intraparenchymal injections targeting a small, anatomically precise region of the midbrain. Their contrast highlights the core challenge of AAV delivery to the human brain: achieving the appropriate balance of vector dose, delivery efficacy, and target tissue volume, a triad of parameters whose interdependencies have shaped every translational strategy to date.

Delivering AAVs to the central nervous system (CNS) is impeded by the blood–brain barrier (BBB), which prevents efficient transduction by most naturally occurring serotypes. As a result, clinical trials have diversified across three primary routes of administration, each representing a distinct position in the dose–efficacy–coverage landscape. Direct intraparenchymal injection offers the highest local delivery efficiency and the most reliable transduction of deep brain nuclei, but it is inherently invasive and limited in the volume of tissue that can be targeted, often requiring multiple surgical trajectories for broader coverage, despite the development of intraoperative MRI (iMRI)-guided convection-enhanced delivery (CED), which enables real-time monitoring and iterative optimization of infusion dynamics. Cerebrospinal fluid (CSF) administration, including intrathecal, intracisternal, and intraventricular routes, provides a less invasive method to distribute vector across the brain's surface and ventricles. However, CSF flows do not naturally penetrate deep parenchyma, resulting in very low transduction efficiencies in deep brain nuclei such as the basal ganglia that are often impacted in neurological disorders. In contrast, intravenous (IV) delivery offers the potential for whole-brain coverage but requires extremely high vector doses to overcome the BBB, with several clinical programs reporting dose-limiting peripheral toxicities in the liver, sensory ganglia, or complement system at the levels necessary for CNS transduction. BBB-penetrant capsids, while enabling broader brain transduction, introduce concerns related to global brain transgene expression and BBB disruption, which may not be desirable or safe for many payloads and disease contexts. Recently, the first patient dosed in a clinical trial of STXBP1 gene therapy for a developmental and epileptic encephalopathy (NCT06983158), which used a single IV dose of a next generation AAV capsid with efficient BBB penetration, died shortly after dosing. While the exact factors involved remain to be elucidated, a recent update from the sponsor indicated that the cause of death was cerebral edema, thus the safety and efficacy of this general approach is not settled. Together, these routes outline the fundamental trade-offs that have constrained clinical translation: invasive approaches deliver efficiently but locally, while non-invasive approaches can reach larger volumes only at doses that challenge safety margins.

Focused ultrasound (FUS) offers a promising complementary strategy by enabling non-invasive, spatially targeted, and transient increase in the permeability of the BBB[6–9]. When combined with systemically administered microbubbles, low-intensity FUS produces mechanical forces that temporarily widen tight junctions between endothelial cells, allowing macromolecules and viral vectors to extravasate into the brain parenchyma. In animal models, FUS-mediated BBB opening (FUS-BBBO) has significantly enhanced AAV transduction efficiency while reducing required systemic doses. Clinically, FUS-BBBO has been demonstrated to be safe and repeatable in more than 30 trials[10] for the delivery of chemotherapies, antibodies, and enzymes—yet notably, no clinical trial has yet evaluated FUS for AAV delivery.

This review will contextualize the emerging capabilities of FUS-mediated AAV delivery within the broader landscape of existing clinical routes of administration, focusing on the interplay between dose, delivery efficacy, and brain volume coverage. For the foreseeable future, multiple strategies for gene therapy delivery will continue to occupy different regions of this landscape, each suited to the distinct anatomical and biological demands of specific neurological diseases—much as Zolgensma and Kebilidi exemplify two extremes of current clinical practice. Our central argument is that FUS-BBBO has the potential to define a new position in this landscape: one that combines the non-invasiveness of systemic administration with the delivery efficiency and spatial precision typically achievable only with direct intraparenchymal injection. Preclinical and clinical FUS studies have already demonstrated BBB opening over volumes comparable to, or exceeding, those targeted by direct injection. The key advances now needed are development of a clinically feasible FUS treatment protocol for safe and effective delivery of AAVs over large, disease-relevant brain regions and the development of AAV capsids specifically engineered for FUS-mediated transport to improve delivery efficacy. Together, these innovations could reduce vector dosing, enhance access to deep or distributed brain regions, and ultimately expand the therapeutic reach of gene therapies whose success is currently limited by delivery constraints.

## Current clinical landscape for AAV delivery to the brain

AAV-based gene therapies have now been evaluated across a broad spectrum of neurological diseases, reflecting both the versatility of the vector platform and the diverse anatomical and biological challenges posed by different disorders. Clinical programs have included single-gene disorders such as aromatic L-amino acid decarboxylase (AADC) deficiency, Huntington's disease, spinal muscular atrophy, metachromatic leukodystrophy, and various lysosomal storage disorders (e.g., Batten disease, MPS I–III), as well as more complex or polygenic diseases such as Parkinson's disease, Alzheimer's disease, amyotrophic lateral sclerosis, multiple system atrophy and epilepsy. These indications differ markedly in how much corrected gene expression is required to achieve therapeutic benefit: enzyme-deficient lysosomal disorders may need only a modest fraction of cells restored to physiological enzyme levels, whereas neurodegenerative diseases with circuit-level dysfunction likely require substantially higher transduction efficiencies distributed across large and heterogeneous brain regions. These differences have shaped the selection of delivery routes used clinically. For this review, we have collected data from 72 trials targeting CNS indications with an AAV-based therapy and the many preclinical studies that supported these trials (see Supplementary Materials). They are roughly evenly divided among direct intraparenchymal injection, administration into CSF spaces (intrathecal or intracisternal), and high-dose intravenous delivery, underscoring both the absence of a single optimal route and the need for disease-specific strategies. **Figure 1** provides a schematic overview of these different routes of administration that will be discussed in this review. The IV category has been split into those using naturally occurring AAV serotypes (such as AAV9) that are only very weakly permeable to the BBB, those using AAVs that have been genetically engineered to be more BBB-penetrant and those adding FUS-BBBO.

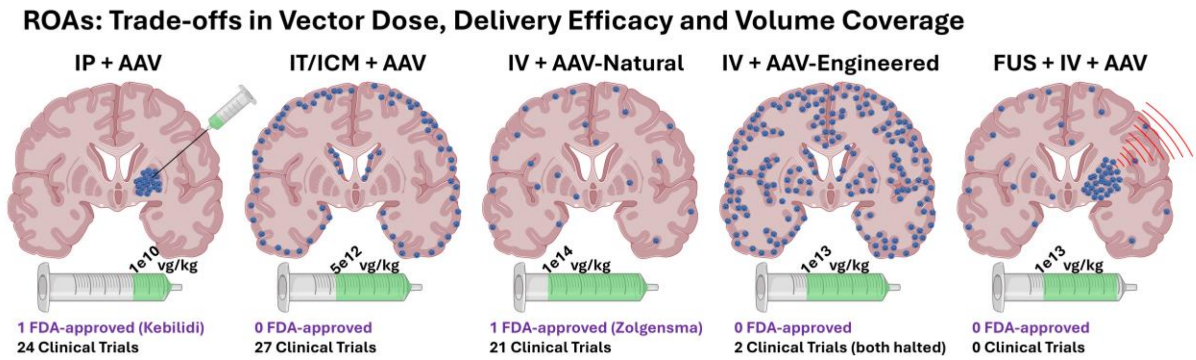


**Figure 1.** Schematic of trade-offs in AAV dose, delivery efficacy and volume coverage for the five routes of administration considered in this review. IP = intraparenchymal; IT = intrathecal; ICM = intra-cisterna magna; IV = intravenous.

*Vector dose*

Among all parameters that define an AAV-based gene therapy, vector dose is one of the most consequential and most variable across clinical trials. As with any therapeutic modality, the fundamental goal is to deliver a dose sufficient to achieve meaningful efficacy while avoiding systemic or organ-specific toxicity. However, unlike small molecules or biologics, AAV dosing spans an exceptionally wide dynamic range in clinical practice. Across CNS-directed studies, doses administered to patients have differed by more than five orders of magnitude on a per-kilogram basis, from approximately 1e9 to 1e14 vg/kg. When accounting for differences in patient body weight—such as comparing a 10-kg infant to a 100-kg adult—the total number of vector genomes administered can vary by nearly seven orders of magnitude, a range unmatched by virtually any other therapeutic platform. This extraordinary variability reflects the distinct challenges posed by different routes of administration and disease targets, as well as the incomplete understanding of how dose interacts with biodistribution, transduction efficiency, and toxicity in the human CNS.

**Figure 2** plots the reported doses used across all CNS-directed AAV trials, split by route of administration[4,5,11–78]. For trials with multiple arms, different doses are reported as separate points. AAV dosing via intraparenchymal injection and intra-CSF delivery are generally not based on body weight calculations (they are based on brain target size and CSF volume, respectively). However, to allow comparison across studies which include systemic dosing in humans, non-human primates and rodents, throughout this review we've normalized all dosing to vg/kg. Trials reporting dose in vg were scaled to vg/kg based on an assumed weight given the age range of the patients in the trial as follows: 0 – 2 yrs = 10 kg; 3 – 15 yrs = 40 kg; 16+ yrs = 70 kg. All trials in the IV category used a naturally occurring AAV serotype. Several trials from the 72 collected did not report the dose(s) used and were therefore not included in this chart. As expected, the dose used increases substantially moving from more invasive to less invasive routes of administration. IP doses are spread over the largest range, reflecting a combination of both the selected dose and the volume of brain tissue to be covered. AAVs administered into CSF spaces cluster around the high-e12 to low-e13 vg/kg range. For IV administration, most trials have clustered around 1e14 vg/kg. The one outlier at 1.6e12 vg/kg is from the low dose cohort in a three-arm trial with escalating doses[78].

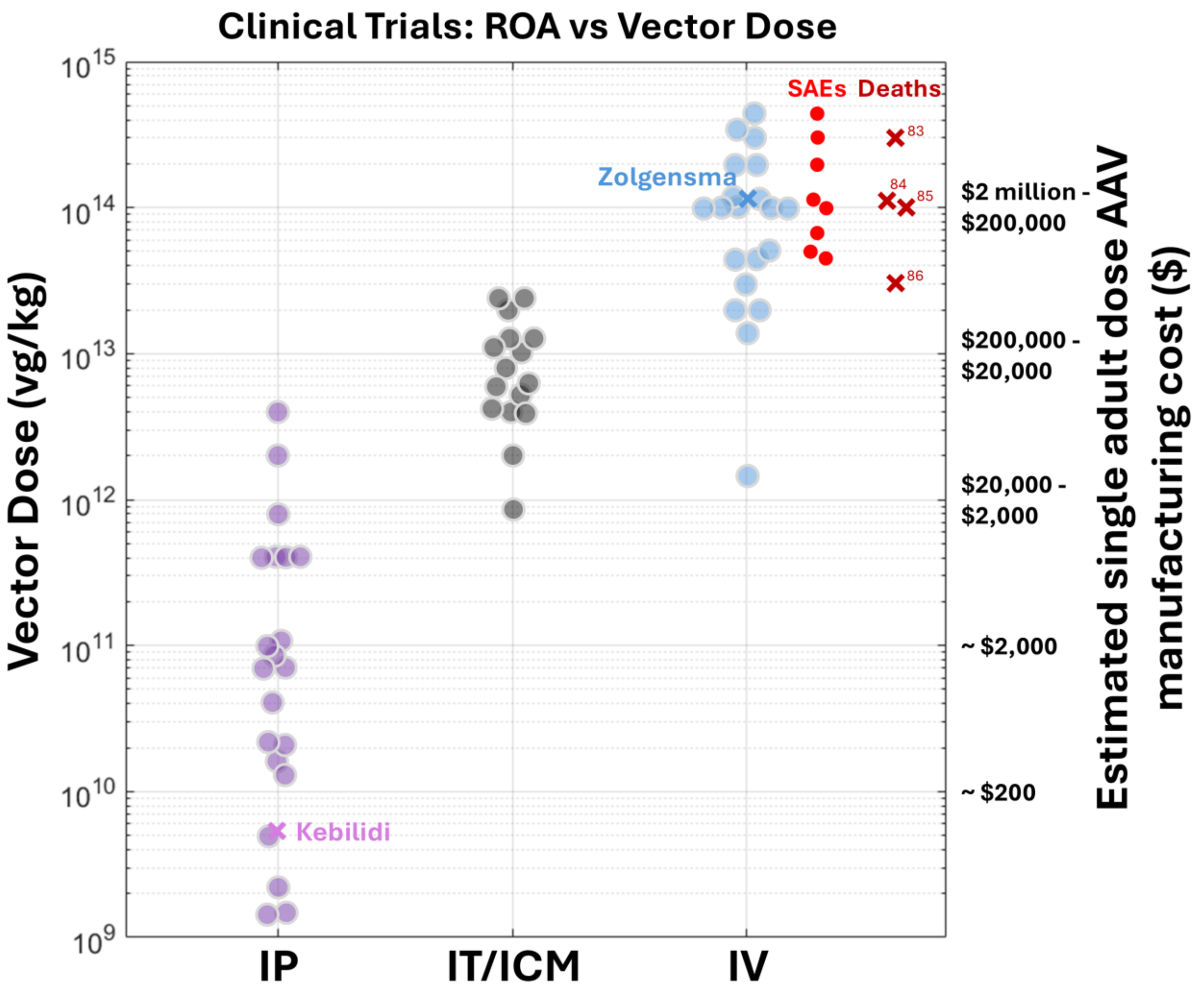


**Figure 2.** Reported AAV doses, in vg/kg, used in clinical trials for IP, IT or ICM and IV administration. For IV-based trials, reported severe adverse events (SAEs) are plotted for the dose administered at which they occurred and reported deaths are plotted for the dose administered at which they occurred, with the numbers indicating the references to the reports. High end single dose manufacturing costs are based on assumptions of $2,000 cost per 1e13 vg of vector and administering to a 100 kg adult patient.

The primary factor constraining the upper limit of AAV dose is toxicity. For IV administrations with naturally occurring AAV serotypes, the principal dangers have been peripheral — most notably hepatotoxicity, dorsal root ganglion (DRG) toxicity, immune activation (complement, inflammation), and other organ-level effects rather than obvious neurotoxicity[79,80]. For example, DRG pathology appears to be dose-dependent and has emerged as a recurring concern with high-dose systemic delivery[81,82]. In a limited number of cases the immune response has been strong enough to cause death[83–86]. Reported significant adverse events (SAEs) and deaths are plotted on the chart for the dose at which they occurred[65,80,87–92]. While most SAEs and deaths occurred for doses at or above 1e14 vg/kg, there are a few instances of serious issues at doses in the mid- to high-e13 vg/kg range. Most recently, a death occurred in a trial using a proprietary engineered AAV capsid, Capsida Biotherapeutics's CAP-002 for STXBP1 developmental and epileptic encephalopathy. While the trial does not report the doses used in patients, the non-human primate studies justifying the trial were done at 2.5e13 vg/kg and 4e13 vg/kg[93]. The first patient dosed in that trial died, prompting a voluntary hold of the study. An autopsy confirmed that the cause of death was cerebral edema, but the underlying cause of the edema remains unclear. Whether the death was directly caused by the AAV infusion, the incident underscores a potential concern: as newer, more efficient capsids achieve greater penetrance levels throughout the entire brain, "on-target" brain-related toxicity — previously less evident — may become a limiting factor.

In addition to the biological trade-off between efficacy and safety, manufacturing cost constitutes a second, practical constraint that may influence which AAV delivery strategies prove commercially viable and broadly deployable. Because AAV manufacturing is laborious, low-yield, and requires stringent purification and quality control, production cost tends to scale with the total number of vector genomes per dose. The right side of the y-axis in **Figure 2** shows a rough estimate of the cost to manufacture the AAVs to dose a single patient. The high-end estimate is based on the assumption that 1e13 vg of vector costs $2,000 to manufacture and is being administered to a 100 kg adult patient. For example, a 1e14 vg/kg dose in a 100 kg patient would require 1e16 vg of vector, requiring 1,000x of the $2,000 cost = $2 million. Clearly, many factors will affect this estimate, from patient weight to the actual cost of producing large batch AAVs that meet strict good manufacturing practice regulatory standards. While the actual costs for a given dose may vary by as much as an order of magnitude, it is clear that trials requiring high systemic doses will incur manufacturing costs orders of magnitude higher than low-dose intraparenchymal or CSF-targeted protocols.

*Volume coverage*

A second major consideration in the design of CNS-directed AAV therapies is the volume of brain tissue that must be effectively covered to achieve clinical benefit. Unlike vector dose—which determines how much AAV reaches the brain—volume coverage determines where it reaches and how extensively. For quantitative comparison of volume coverage across the different routes of administration, we assume IV administration of AAVs achieves whole-brain coverage, with the caveats that, in practice, IV-delivered vectors do not transduce all regions equally, with significant variability arising from differences in vascular density, BBB properties, and serotype-specific tropism. This section focuses on the two modalities designed for regional or focal targeting: IP injection and FUS-BBBO. For many neurological diseases, the therapeutic goal is to correct pathology in a specific subregion where the biology of the disorder is most concentrated—often in deep brain structures such as the thalamus, basal ganglia, or substantia nigra. The feasibility of reaching these clinically critical targets, and doing so at sufficient volume and density, is a defining factor in evaluating and comparing both IP and FUS-based delivery strategies. The FDA approval for Kebilidi and the exciting results from uniQure's Phase I/II clinical trial delivering their HTT-lowering AMT-130 therapy to Huntington's disease patients by direct injection to the striatum, achieving a ~75% reduction in disease progression at 36 months[94], demonstrates that targeting critical brain structures, even when they comprise only a small fraction of total brain volume, is sufficient to meaningfully slow disease progression.

Once sufficient coverage of the disease-relevant region is achieved, limited spatial exposure may itself become advantageous from a safety perspective. Systemic, whole-brain delivery exposes all neural and glial cell populations to AAV capsids and vector genomes, which are known to trigger innate immune responses, microglial activation, and astrocytic gliosis, particularly at higher doses. In contrast, focal strategies such as IP injection or FUS-BBBO confine vector exposure to selected regions, potentially limiting widespread neuroinflammatory signaling. Moreover, restricting delivery to anatomically defined targets may reduce the risk of off-target transgene expression in cell types where the gene is not normally expressed, thereby mitigating unintended toxicity while preserving therapeutic benefit.

**Figure 3** summarizes the reported volume coverage achieved to date by direct intraparenchymal (IP) injection—separated into preclinical and clinical studies—and by FUS-BBBO in clinical trials[5,11–15,19,21,23,25,29,95–119]. As with AAV dose, the y-axis spans several orders of magnitude, from the volume of a single injection covering 0.1 cubic centimeters (cc) to the size of the whole human brain at roughly 1,400 cc. The FUS-BBBO data derive not from AAV trials, which have not yet entered the clinic, but from studies evaluating the safety, feasibility, and drug-delivery capabilities of ultrasound-mediated BBB opening using small-molecule chemotherapies, antibodies, or enzymes. For FUS-BBBO trials, reported coverage volumes are based on MRI measurements of gadolinium enhancement on T1-weighted images. Although gadolinium is much smaller than an AAV particle, multiple animal studies have demonstrated that the spatial distribution of gadolinium following FUS-BBBO strongly correlates with AAV extravasation and transduction, validating its use as a surrogate imaging marker for estimating potential AAV coverage. For the preclinical and clinical IP data, volumes based on MRI measurements of gadolinium enhancement are used when reported. As most studies did not report this information, volumes for these studies were determined by scaling the reported injected volume, $V_i$, to a volume of delivery, $V_d$, based on the validated factor of $V_d/V_i = 3$ [120].

The largest volume coverage achieved by direct IP injection in a human is approximately 7 cc, which corresponds to the volume of the bilateral human putamen (≈6.4–8.9 cc). Importantly, even much smaller coverage volumes have yielded meaningful clinical benefit, demonstrating that focal correction in a strategically chosen deep brain structure can be sufficient for therapeutic effect. For example, the AAV2-bsaed Kebilidi is administered through four intraparenchymal infusions (two per putamen) with a total infusion volume of 0.32 ml, translating to an estimated 0.96 cc of tissue coverage—a small fraction of the basal ganglia (20 cc), yet sufficient to drive substantial and durable clinical improvement. A similar pattern is seen with uniQure's AMT-130 (AAV5) program for Huntington's disease, where patients receive six injections (three per hemisphere) targeting the caudate and putamen. Although the exact infusion volume per site is not publicly disclosed, expected values of 0.3–0.5 cc per injection would produce a total coverage of approximately 1.8–3.0 cc, again representing focal but clinically relevant modulation of a deep brain circuit.

Clinical FUS-BBBO studies have demonstrated a wide range of achievable coverage volumes, spanning approximately 0.1 cc to 40 cc, depending on the therapeutic target and the number of sonication locations used. Each individual FUS exposure generates a focal region of BBB opening roughly 2 × 2 × 2 mm in size, and larger coverage volumes are created by tiling multiple targets sequentially to cover a user-defined volume. Early human trials focused on establishing safety and feasibility by opening the BBB in small cortical regions, often limited to a handful of adjacent sonication sites. To date, more than 30 clinical trials have evaluated FUS-BBBO, with approximately two-thirds focused on oncology applications—primarily glioblastoma—and the remainder targeting neurodegenerative diseases such as Alzheimer's disease, Parkinson's disease, and amyotrophic lateral sclerosis. These early studies established the safety and repeatability of BBB opening in humans, and more recent trials have begun to demonstrate both delivery and therapeutic efficacy. Notably, FUS-BBBO has been shown to enhance delivery of antibodies such as aducanumab in Alzheimer's disease patients[99], and a recent multi-center Phase 1/2 trial in glioblastoma reported an approximately 10-month survival benefit when FUS-BBBO was added to standard-of-care temozolomide[121]. More recent studies have expanded both the anatomical breadth and depth of treatment, targeting larger cortical territories for Alzheimer's disease and, notably, deeper structures such as the substantia nigra in exploratory trials for Parkinson's disease. The largest reported FUS-BBBO volumes, up to 40 cc, now exceed those achieved by direct intraparenchymal injection and are comparable to, or greater than, the combined bilateral volume of the human basal ganglia.

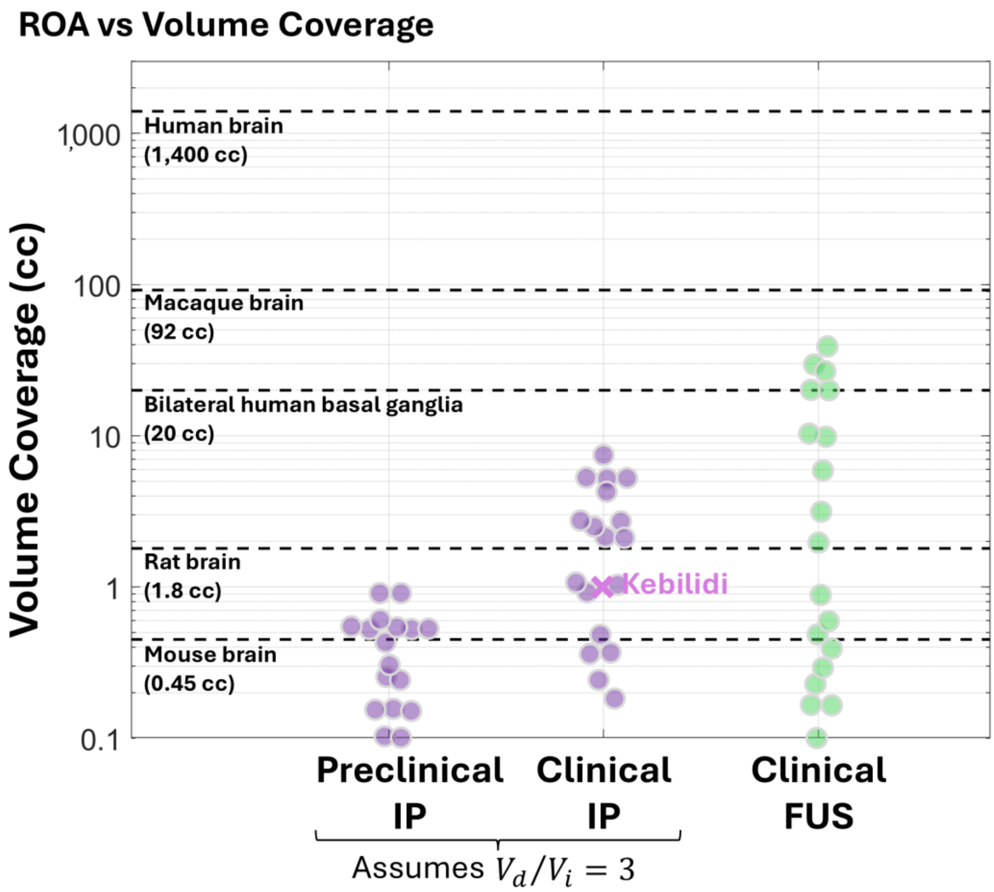


**Figure 3.** Preclinical IP studies and IP clinical trials reporting either the volume of therapy coverage based on MRI or the volume of AAV therapy injected ($V_i$), which was then scaled to volume coverage ($V_d$) based on the assumption that $V_d/V_i = 3$ . Volume coverage for FUS trials are based on contrast MRI measurements.

*Delivery efficacy*

Unlike vector dose or volume coverage, delivery efficacy—the degree to which administered AAV successfully enters the brain and transduces target cells—cannot be measured directly in humans outside of rare biopsy or postmortem analyses. As a result, our understanding of how efficacy scales with administered dose largely relies on preclinical animal studies, which allow systematic evaluation across the five major routes of administration outlined in Figure 1. In these models, delivery efficacy can be quantified in several complementary ways, including the percentage of cells transduced (typically measured using fluorescent reporter genes such as GFP), the number of viral genomes per cell, or the number of payload transgene copies per cell, each providing a slightly different window into the efficiency of vector entry, genome conversion, and expression. Early work across all routes of administration focused primarily on naturally occurring AAV serotypes, such as AAV9, to establish baseline dose–response relationships and to understand the inherent limitations imposed by BBB transport. More recently, there has been intense effort to engineer AAV capsids with enhanced BBB penetrance for IV administration, yielding a new generation of vectors that achieve substantially higher transduction efficiencies in both mice and non-human primates. FUS-BBBO provides a second mechanism for penetrating the BBB and can be combined with both naturally occurring and engineered AAVs to further improve delivery efficacy.

**Figure 4** summarizes delivery efficacy, measured as the percentage of neurons or cells transduced, as a function of administered AAV dose across the five routes of administration, drawing on data collated from preclinical studies in mice, rats and non-human primates[93,111,114,122–155]. AAV doses for IP and

IT/ICM routes of administration have been scaled to vg/kg as described above. While these data only represent the subset of preclinical studies that have reported outcomes in terms of percent cells transduced, they are sufficient to establish trends across the different routes of administration. As expected, intraparenchymal (IP) injection achieves high levels of neuronal transduction even at very low doses, reflecting the advantage of bypassing the BBB altogether; recent work combining IP delivery with engineered capsids has further enhanced both efficiency and parenchymal spread[155]. Intrathecal and intracisternal (IT/ICM) routes support modest to strong delivery though delivery efficiency is highly region-dependent and generally requires higher doses in the range of 5e12 to 5e13 vg/kg, reflecting the need for vectors to diffuse from CSF spaces into parenchymal tissue. In contrast, intravenous (IV) administration with naturally occurring AAV serotypes shows the weakest delivery efficacy at 1–10% neuronal transduction, underscoring the limited innate BBB penetrance of these vectors. However, the development of engineered BBB-permeant AAV capsids has transformed the IV landscape: several of these vectors now achieve ~50% transduction in deep structures such as striatum and up to 90% in cortical regions at doses near 1e13 vg/kg, representing a remarkable improvement over natural serotypes. To date, FUS + AAV studies have primarily paired FUS-BBBO with naturally occurring AAVs administered at doses well below the threshold required for self-penetrant IV delivery. These studies nonetheless show clear enhancements in delivery, though relatively few report neuron-specific transduction rates; those that do generally observe ~5–10% neuronal transduction at these low doses, suggesting substantial room for improvement as both FUS treatment parameters and capsid design become optimized for this modality. There are only a few instances of pairing FUS with engineered AAV capsids, which results in a similar jump in delivery efficacy compared to natural capsids. However, these studies have only been done in mice to date and must be validated in primates.

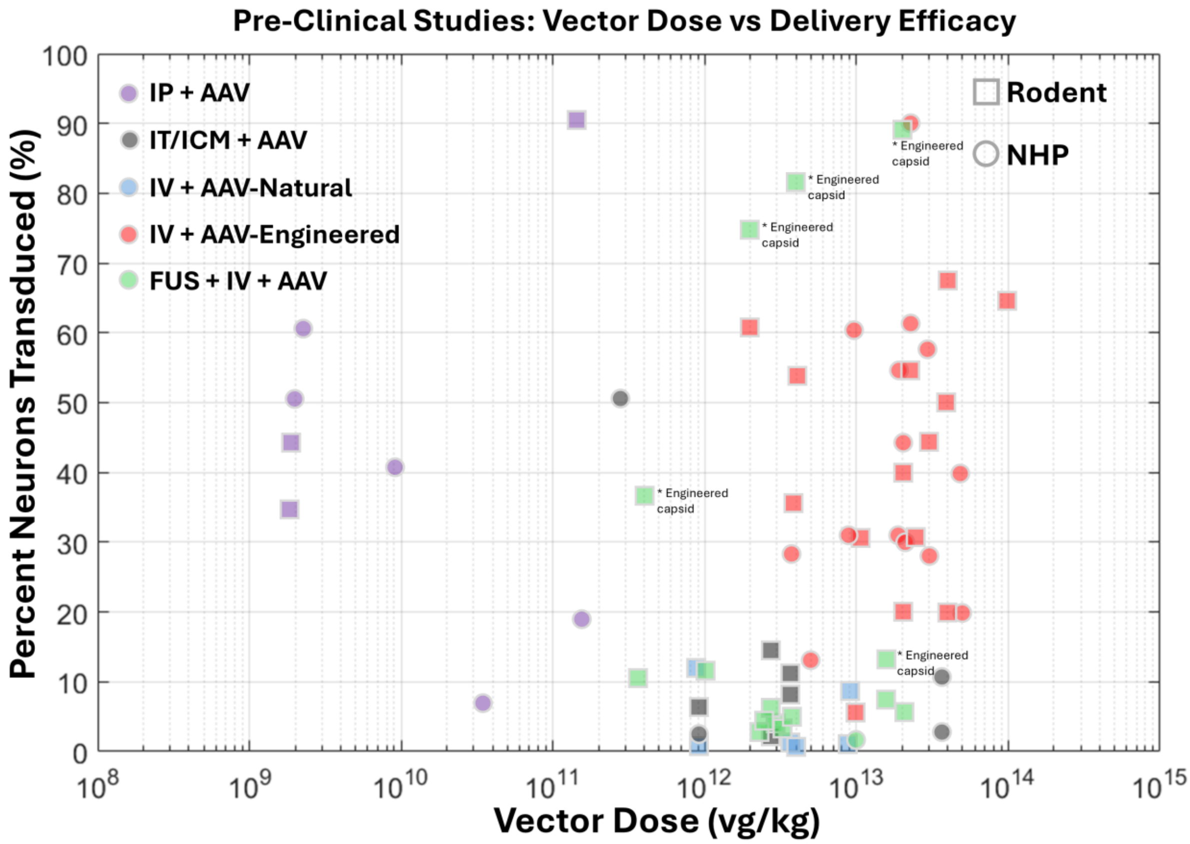


**Figure 4.** Preclinical studies reporting percentage of neurons or total cells transduced as a function of AAV dose for different routes of administration. Rodent vs NHP studies are distinguished by circles vs squares. As AAV dosing for IP and IT/ICM delivery are generally not based on body weight calculations we have scaled the reported doses from vg to vg/kg for appropriate comparison across animal models. Studies reporting dose in vg were scaled

to vg/kg based on an assumed weights of 25g for mice, 300g for rats and 5 kg for NHPs. FUS + IV + AAV studies used naturally occurring AAV serotypes unless noted as using an engineered vector.

## FUS in the existing landscape: current status and next key advances

The individual data points from the 72 clinical trials and many preclinical studies that went into Figures 2 - 4 are summarized in **Figure 5**, showing the approximate regimes that the different routes of administration occupy in the dose, delivery and volume coverage landscape. As noted previously, the two FDA-approved therapies, Zolgensma and Kebilidi, operate in very different spaces in this landscape, differing by several orders of magnitude in both dose and volume covered. In this section, we argue that FUS-BBBO, when paired with an AAV vector specifically optimized for ultrasound-mediated transport, has the potential to unlock a new region of this landscape, situated between these two approved therapies. A clinically meaningful FUS-BBBO treatment paradigm might reasonably cover ~20 cc of a targeted brain region, utilize an intermediate systemic dose on the order of 1e13 vg/kg, and achieve delivery efficiencies approaching 40% of cells transduced—a combination not currently accessible with existing delivery routes. Such an approach would be attractive for several reasons: it would extend coverage to deep or distributed structures without the invasiveness and limited spread of direct intraparenchymal injection; it would reduce the systemic dose requirements, thereby improving both safety and commercial feasibility; and it would confine vector delivery to the most clinically relevant brain regions, that only comprise ~2% of total brain tissue, to limit the possibility of brain-related toxicity.

However, realizing this capability will require further advances in FUS-mediated delivery efficiency, which remains below what is needed for many disease targets. This will likely be achieved through a combination of optimized FUS treatment parameters to maximize BBB permeability over large volumes and AAV capsid engineering specifically tuned to exploit the transient BBB opening and mechanical transport pathways created by FUS, both of which are subjects of ongoing research.

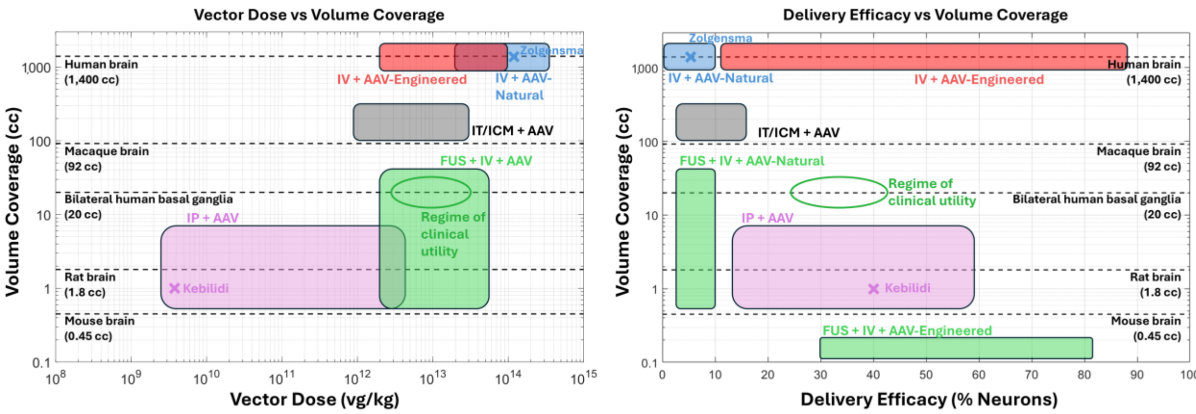


**Figure 5.** The data from Figures 2 – 4 summarized in two plots to show the full landscape of dose, delivery efficacy and volume coverage for different routes of administration. For Delivery Efficacy, FUS + IV + AAV has been split into AAV-Natural and AAV-Engineered given both the large difference in achievable delivery efficiency and that FUS + IV + AAV-Engineered has not yet been validated in a large brain. The outline regime of clinical utility represents our goal for the field to achieve 20cc volume coverage and ~40% neuronal transduction at ~1e13 vg/kg systemic dose.

### *FUS treatment parameters*

FUS-BBBO is achieved by a short burst ultrasonic pressure wave acting on microbubbles circulating in the vasculature, causing them to cavitate and thereby exert mechanical forces on vessels walls. The primary mechanism of BBB permeability is weakening of tight junctions between endothelial cells that allows for passive diffusion of agents from the bloodstream into the brain parenchyma[156,157]. There is also some evidence that FUS increases the presence of cellular vesicles[158], which may be a plausible secondary mechanism of penetrance for AAVs. Safe and effective BBB disruption has been validated in animals and patients, with a typical parameter set consisting of ~500 kHz ultrasound frequency, ~0.3 MPa ultrasound pressure, 20 µl/kg microbubble dose and 10 ms ultrasound bursts repeated at a pulse repetition frequency (PRF) of once per second (1 Hz) for 120 seconds. After treatment, BBB opening can be confirmed using gadolinium-based contrast MRI which shows hyperintense signal in regions where the gadolinium has extravasated from the vasculature. While the BBB is maximally permeable immediately after FUS treatment, it remains open for several hours and typically fully re-closed by 24 hours post-treatment[159], allowing AAVs to be administered after the application of ultrasound and subsequent confirmation of successful BBB disruption on MRI[158].

Optimizing these parameters to maximize AAV delivery remains an active area of investigation. The parameter space is large—spanning acoustic frequency, burst duration, duty cycle, pressure amplitude, microbubble formulation and dose, and the geometric pattern of sonication—but accumulating evidence suggests that pressure amplitude and microbubble dose are likely the dominant drivers of the magnitude of BBB opening. Achieving the optimal balance is challenging, as higher pressures and bubble concentrations can increase permeability but also raise the risk of vascular injury and instigation of an acute inflammatory response. To navigate these trade-offs, modern systems increasingly incorporate real-time monitoring of microbubble acoustic emissions, enabling feedback-controlled sonication in which pressure is dynamically adjusted to maintain stable cavitation while avoiding harmful bubble collapse.

A central challenge in treatment parameter optimization is achieving large-volume coverage within a clinically reasonable treatment time. Modern clinical systems typically use ~1,000-element hemispherical phased-array transducers that produce a natural focal spot of roughly 2 × 2 × 2 mm. While this small focus is advantageous for precise feedback control, building up large volume BBB disruption by sequential sonication of individual targets is infeasible: adding single 8-$mm^3$ volume elements every two minutes would require many hours to cover tens of cubic centimeters of tissue. The key to scaling FUS-BBBO lies in exploiting the capabilities of phased arrays, which can electronically steer the focus and deliver ultrasound energy to multiple targets within a single pulse repetition frequency (PRF). Current systems can sonicate 32 different targets per PRF, and in principle, with 10-ms bursts at 1 Hz, could support up to 100 targets per PRF. These capabilities dramatically accelerate volumetric coverage by allowing the system to tile dozens of focal spots in parallel rather than sequentially. In addition, phased arrays enable acoustic holography, in which phase and amplitude modulation across array elements create extended or sculpted focal geometries[160]. Such holographic patterns can deposit energy over larger contiguous regions, reducing the number of discrete targets needed to fill a therapeutic volume. **Figure 6A** shows a schematic example of 2x2x2 mm targeting in a mouse brain and the much larger human brain. **Figure 6B** plots the achievable volume coverage as a function of time for three different treatment parameter sets. Treatment time scales linearly with the number of targets, the number of bursts delivered per target, and the number of targets sonicated per PRF, such that some combination of parallel sonication, burst-number optimization, and holographic focusing can reduce treatment times sufficiently to achieve ~20 cc of BBB opening within a clinically feasible timeframe.

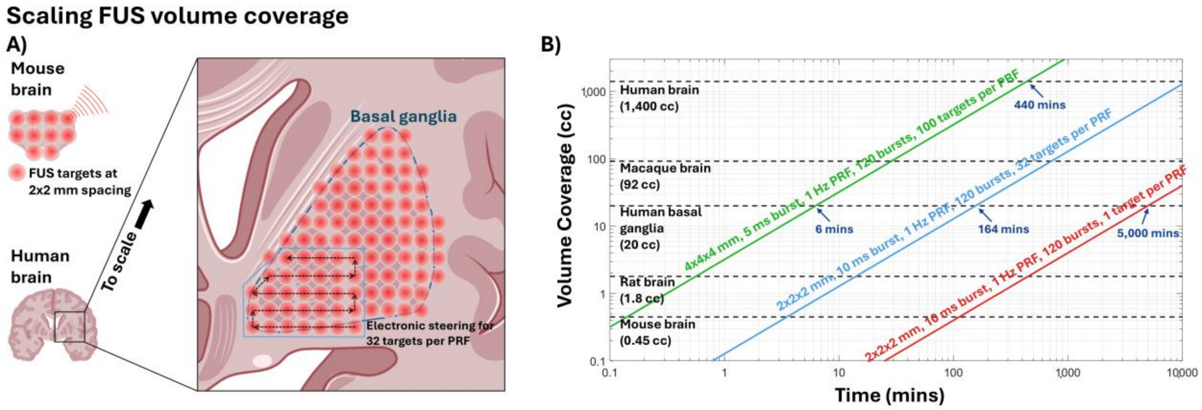


**Figure 6.** A) Schematic representation of FUS targets at 2x2 mm spacing in a single plane in the mouse and human brain. B) Three examples of FUS treatment parameter sets and the achievable volume coverage as a function of time. Preclinical studies are typically done under the red set of parameters which is prohibitively time consuming to achieve meaningful volume coverage in a human patient. A combination of increasing the number of sonicated targets per PRF, reducing the number of bursts per target and increasing the FUS focal volume to reduce the number of total targets needed can lead to substantial reductions in treatment time.

*FUS-optimized AAV vectors*

While treatment parameter optimization will help improve delivery efficacy to some extent, the largest gains will likely come from discovering or specifically engineering an AAV vector that exploits the unique physiological changes to the BBB induced by FUS. Several research groups have begun to pursue this, and the early results are highly encouraging. **Figure 7** shows one example from our group of the gains in delivery efficacy that can be achieved when combining FUS with an engineered AAV vector compared to FUS + AAV9[161]. FUS combined with the capsid AAV.CPP16[135] was able to achieve delivery efficacies of 70% to 90% in the targeted mouse cortex at very low systemic doses of 2e12 to 2e13 vg/kg. Another group has seen similar improvements over FUS + AAV9 when using a capsid developed through a directed evolution approach, in which a large library containing randomized peptide insertions was screened *in vivo* under FUS-BBBO conditions to select for variants with enhanced penetration and transduction[132]. The reported delivery efficiency for this capsid was in the range of 20% to 40% of neurons transduced at the very low dose of 4e11 vg/kg. A major caveat, however, is that these FUS-optimized capsids have been validated only in mouse models; reproducing these enhancements in non-human primates—where differences in BBB structure, immune responses, and tropism often diminish performance—is a critical next step before translation to human trials. Recently, another group demonstrated successful delivery to the non-human primate caudate and putamen using FUS-BBBO combined with two different engineered AAV capsids, AAV2-HBKO and AAV.GMU01[162]. Notably, this study used ICM injection as the route of administration, as a strategy to combine the cortical distribution achievable with CSF-based delivery and the deep-brain targeting of FUS-BBBO.

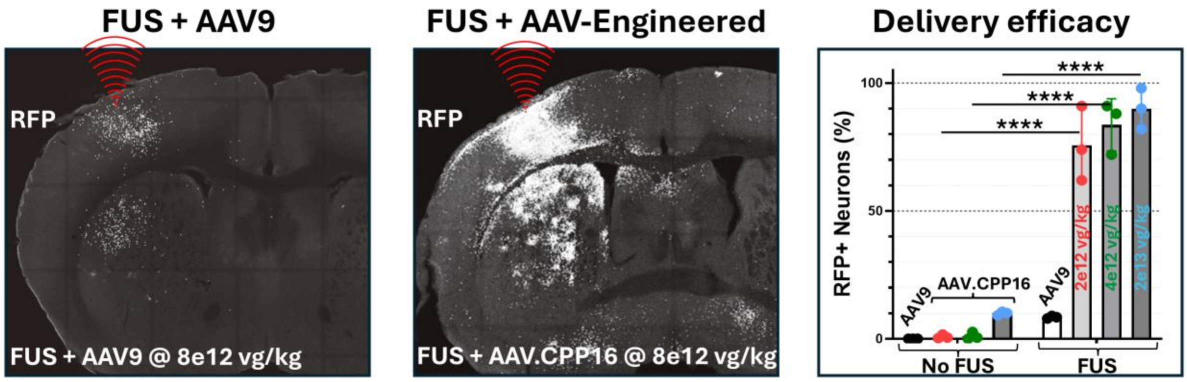


**Figure 7.** Preclinical example of FUS-BBBO combined with an engineered AAV vector, AAV.CPP16, for improved delivery over what can be achieved with FUS + AAV9. FUS-BBBO was applied to the left cortex and striatum of wild-type BALB/C mice with either AAV9-RFP or AAV.CPP16-RFP administered by tail vein injection immediately after. Examples images show RFP expression at 3 weeks post treatment for 8e12 vg/kg dose. Plot shows quantification of neuronal transduction for AAV9 at 8e12 vg/kg and AAV.CPP16 at doses of 2e12, 4e12, and 2e13 vg/kg, in both FUS-targeted (FUS) and non-targeted (No FUS) brain regions.

## Summary and Conclusions

Across two decades of clinical and preclinical research, it has become clear that no single route of administration fully solves the central challenge of delivering AAVs to the human brain. Each occupies a distinct position within the broader dose–efficacy–volume coverage landscape, with IP injection providing focal but invasive delivery, IT/ICM routes offering moderate coverage at intermediate doses but lack of deep structure penetration, and IV delivery enabling diffuse, whole-brain access but with dose-limiting toxicities and narrow therapeutic index. Focused ultrasound–mediated BBB opening (FUS-BBBO) offers a compelling path toward a new regime in this landscape—one that combines the non-invasiveness of systemic administration at a sub-toxic dose level with the spatial targeting of IP delivery and the ability to achieve selective coverage of the brain regions most affected by disease. The technical foundations for such an approach are rapidly advancing: modern phased-array systems now support multi-target and holographic sonication strategies; real-time acoustic feedback enables safe and consistent BBB modulation; and an emerging focus on developing FUS-optimized AAV capsids promises to dramatically improve delivery efficiency at clinically reasonable doses. Taken together, the essential pieces needed to achieve low systemic dose (~1e13 vg/kg), broad coverage (~20 cc), and high neuronal transduction rates (~40%) are increasingly in place—what remains is to optimize these components specifically for AAV delivery and to validate their combined performance in large-animal models. Success on these fronts would provide a clear translational pathway toward first-in-human trials and establish FUS-BBBO as a transformative new modality for CNS gene therapy.

## Supplementary Materials

List of references used for Figures 2 – 4.

### Fig 2: Vector dose in clinical trials

| ROA | Author / Sponsor | Journal | Year | PMID / NCT |
|---|---|---|---|---|
| IC | Chien | Lancet Child Adolesc Health | 2017 | 30169182 |
| IC | Pearson | Nat Commun | 2021 | 34253733 |
| IC | Rafii | Alzheimers Dement | 2014 | 24411134 |
| IC | Souweidane | J Neurosurg Pediatr. | 2011 | 20672930 |
| IC | Sondhi | Sci Transl Med | 2020 | 33268510 |
| IC | Leone | Sci Transl Med | 2012 | 23253610 |
| IC | Estevez-Fraga | J Huntingtons Dis | 2024 | 38489195 |
| IC | Regeneron | | | NCT03580083 |
| IC | Regeneron | | | NCT03566043 |
| IC | Bugiani | Ann Clin Transl Neurol | 2023 | 37165777 |
| IC | Tardieu | Hum Gene Ther. | 2014 | 24524415 |
| IC | Gougeon | Front Immunol. | 2021 | 34040605 |
| IC | Brain Neurotherapy | | | NCT04680065 |
| IC | Christine | Neurology | 2022 | 34649873 |
| IC | Bankiewicz | | | NCT07267065 |
| IC | LeWitt | Lancet Neurol | 2011 | 21419704 |
| IC | Heiss | Mov Disord | 2019 | 31145831 |
| IC | Bartus | Neurology | 2013 | 23576625 |
| IC | Marks | Lancet Neurology | 2010 | 20970382 |
| IC | Eichler | Nat Med | 2025 | 40817303 |
| IC | Corti | Mol Ther Methods Clin Dev | 2023 | 37601414 |
| IC | Yilmaz | J Inherit Metab Dis | 2020 | 32944950 |
| IC | Xinzhi BioMed | | | NCT07195825 |
| IC | Hoffmann-La Roche | | | NCT06826612 |
| IT/ICM | Eichler | Nat Med | 2025 | 40817303 |
| IT/ICM | Gemma Biotherapeutics | | | NCT04771416 |
| IT/ICM | REGENXBIO | | | NCT04571970 |
| IT/ICM | Sevigny | Nat Med | 2024 | 38745011 |
| IT/ICM | Myrtelle Inc | | | NCT04833907 |
| IT/ICM | Prevail | | | NCT04127578 |
| IT/ICM | Prevail | | | NCT04411654 |
| IT/ICM | LYSOGENE | | | NCT04273269 |
| IT/ICM | Mueller | N Engl J Med | 2020 | 32640133 |
| IT/ICM | Alcyone Therapeutics | | | NCT03770572 |
| IT/ICM | Emily de los Reyes | | | NCT02725580 |
| IT/ICM | Benjamin Greenberg | | | NCT04737460 |
| IT/ICM | Novartis | | | NCT03381729 |
| IT/ICM | Bharucha-Goebel | N Engl J Med | 2024 | 38507752 |
| IT/ICM | SwanBio Therapeutics | | | NCT05394064 |
| IT/ICM | Strauss | Nat Med | 2022 | 35715567 |
| IT/ICM | Anupam Sehgal | | | NCT04798235 |
| IT/ICM | Guangzhou Women and Children's Medical Center | | | NCT06856759 |
| IT/ICM | GeneCradle | | | NCT06971094 |
| IT/ICM | SineuGene Therapeutics | | | NCT07169175 |
| IT/ICM | RJK Biopharma | | | NCT06454682 |
| IT/ICM | Jaguar Gene Therapy | | | NCT06662188 |
| IT/ICM | BlackfinBio | | | NCT06948019 |
| IT/ICM | Xinhua Hospital | | | NCT06272149 |
| IT/ICM | Genecombio | | | NCT06772402 |
| IT/ICM | Yongguo Yu | | | NCT06860672 |
| IT/ICM | MavriX Bio | | | NCT07181837 |
| IV | Libella Gene Therapeutics | | | NCT04133454 |
| IV | Forge Biologics | | | NCT04693598 |
| IV | Spur Therapeutics | | | NCT05324943 |
| IV | Mendell | N Engl J Med | 2017 | 29091557 |
| IV | Harmatz | Mol Ther | 2022 | 36299240 |
| IV | Ultragenyx Pharmaceutical | | | NCT02716246 |

| | | | | |
|---|---|---|---|---|
| **IV** | Gougeon | Front Immunol | 2021 | 34040605 |
| **IV** | Lewis | medRxiv | 2025 | 40766118 |
| **IV** | Aspa Therapeutics | | | NCT04998396 |
| **IV** | Mendell | Nat Med | 2025 | 39385046 |
| **IV** | Shieh | Lancet Neurol | 2023 | 37977713 |
| **IV** | Megan Waldrop | | | NCT04240314 |
| **IV** | Solid Biosciences | | | NCT03368742 |
| **IV** | Mendell | JAMA Neurol | 2020 | 32539076 |
| **IV** | Butterfield | Nat Med. | 2025 | 40579547 |
| **IV** | Mendell | Nat Med | 2024 | 38177855 |
| **IV** | Spark Therapeutics | | | NCT04093349 |
| **IV** | Abeona Therapeutics | | | NCT03315182 |
| **IV** | Pfizer | | | NCT04281485 |
| **IV** | Zaidman | Ann Neurol | 2023 | 37539981 |
| **IV** | Smith | Mol Ther | 2023 | 36805083 |

### Fig 3: Volume coverage

| ROA | Author | Journal | Year | PMID |
|---|---|---|---|---|
| **FUS** | Mehta | Radiology | 2021 | 33399511 |
| **FUS** | Mehta | Fluids Barriers CNS | 2023 | 37328855 |
| **FUS** | Lipsman | Nat Commun. | 2018 | 30046032 |
| **FUS** | Rezai | PNAS | 2020 | 32284421 |
| **FUS** | Rezai | N Engl J Med. | 2024 | 38169490 |
| **FUS** | Mainprize | Sci. Rep. | 2019 | 30674905 |
| **FUS** | Anastasiadis | PNAS | 2021 | 34504017 |
| **FUS** | Chen | Sci Adv. | 2021 | 33547073 |
| **FUS** | Meng | Sci Transl Med. | 2021 | 34644145 |
| **FUS** | Carpentier | Sci Transl Med. | 2016 | 27306666 |
| **FUS** | Gasca-Salas | Nat Commun. | 2021 | 33536430 |
| **FUS** | Meng | Mov Disord. | 2022 | 36089809 |
| **FUS** | Abrahao | Nat Commun. | 2019 | 31558719 |
| **Preclinical IC** | Bankiewicz | Exp Neurol. | 2000 | 10877910 |
| **Preclinical IC** | Fiandaca | Neuroimage | 2009 | 19095069 |
| **Preclinical IC** | Rosenberg | Hum Gene Ther Clin Dev | 2018 | 29409358 |
| **Preclinical IC** | Hadaczek | Mol Ther Methods Clin Dev | 2016 | 27408903 |
| **Preclinical IC** | Cederfjäll | Sci Rep. | 2013 | 23831692 |
| **Preclinical IC** | Herzog | Mov Disord. | 2007 | 17443702 |
| **Preclinical IC** | Hadaczek | Mol Ther. | 2010 | 20531394 |
| **Preclinical IC** | Kells | J Neurosci. | 2010 | 20631185 |
| **Preclinical IC** | Johnston | Hum Gene Ther. | 2009 | 19203243 |
| **Preclinical IC** | Rosenberg | Hum Gene Ther. | 2021 | 33380277 |
| **Clinical IC** | Chien | Lancet Child Adolesc Health | 2017 | 30169182 |
| **Clinical IC** | Pearson | Nat Commun. | 2021 | 34253733 |
| **Clinical IC** | Rafii | Alzheimers Dement | 2014 | 24411134 |
| **Clinical IC** | Souweidane | J Neurosurg Pediatr | 2010 | 20672930 |
| **Clinical IC** | Sondhi | Sci Transl Med | 2020 | 33268510 |
| **Clinical IC** | Leone | Sci Transl Med | 2012 | 23253610 |
| **Clinical IC** | Bugiani | Ann Clin Transl Neurol | 2023 | 37165777 |
| **Clinical IC** | Gougeon | Front Immunol | 2021 | 34040605 |
| **Clinical IC** | Christine | Neurology | 2022 | 34649873 |
| **Clinical IC** | Christine | Ann Neurol | 2019 | 30802998 |
| **Clinical IC** | LeWitt | Lancet Neurol | 2011 | 21419704 |
| **Clinical IC** | Marks | Lancet Neurol | 2008 | 18387850 |
| **Clinical IC** | Eichler | Nat Med | 2025 | 40817303 |

### Fig 4. Delivery vs dose

| ROA | Author | Journal / Conference | Year | PMID |
|---|---|---|---|---|
| **IC** | Chandran | Gene Ther | 2023 | 35637286 |
| **IC** | Hadaczek | Mol Ther Methods Clin Dev | 2016 | 27408903 |
| **IC** | Pignataro | Front Neuroanat | 2017 | 28239341 |

| | | | | |
|---|---|---|---|---|
| **IC** | McFarland | J Neurochem | 2009 | 19250335 |
| **IC** | Hadaczek | Mol Ther. | 2010 | 20531394 |
| **IT/ICM** | Chandran | Gene Ther | 2023 | 35637286 |
| **IT/ICM** | Gray | Gene Ther | 2013 | 23303281 |
| **IT/ICM** | Liguore | Mol Ther. | 2019 | 31420242 |
| **IT/ICM** | Chauhan | J Trans Med | 2024 | 39237935 |
| **IT/ICM** | Garza | Gene Ther | 2025 | 39501094 |
| **IT/ICM** | Castle | Sci. Adv. | 2018 | 30443600 |
| **IT/ICM** | Fortuna | Mol Ther Methods Clin Dev | 2025 | 41438872 |
| **IV + AAV-Natural** | Hanlon | Mol Ther Methods Clin Dev | 2019 | 31788496 |
| **IV + AAV-Natural** | Chauhan | J Trans Med | 2024 | 39237935 |
| **IV + AAV-Natural** | Li | Nat Commun | 2024 | 38858354 |
| **IV + AAV-Natural** | Hsu | Plos One | 2013 | 23460893 |
| **IV + AAV-Natural** | Nouraein | Gene Ther | 2024 | 37696982 |
| **IV + AAV-Natural** | Yao | Nat Biomed Eng. | 2022 | 36217021 |
| **IV + AAV-Eng.** | Chan | Nat Neurosci | 2017 | 28671695 |
| **IV + AAV-Eng.** | Lin | Cell Rep. | 2025 | 41175373 |
| **IV + AAV-Eng.** | Stanton | Med. | 2023 | 36417917 |
| **IV + AAV-Eng.** | Chauhan | J Trans Med | 2024 | 39237935 |
| **IV + AAV-Eng.** | Huang | Science | 2024 | 38753766 |
| **IV + AAV-Eng.** | Yao | Nat Biomed Eng. | 2022 | 36217021 |
| **IV + AAV-Eng.** | Deshpande | ASGCT Proceedings | 2025 | -- |
| **IV + AAV-Eng.** | Maura | ASGCT Proceedings | 2025 | -- |
| **IV + AAV-Eng.** | Shi | ASGCT Proceedings | 2025 | -- |
| **IV + AAV-Eng.** | Ye | ASGCT Proceedings | 2025 | -- |
| **IV + AAV-Eng.** | Knoll | ASGCT Proceedings | 2025 | -- |
| **IV + AAV-Eng.** | Moyer | ASGCT Proceedings | 2024 | -- |
| **IV + AAV-Eng.** | Shi | ASGCT Proceedings | 2024 | -- |
| **IV + AAV-Eng.** | Tiffany | ASGCT Proceedings | 2024 | -- |
| **FUS + IV + AAV** | Kofoed | J Control Release | 2022 | 36179767 |
| **FUS + IV + AAV** | Li | Nat Commun | 2024 | 38858354 |
| **FUS + IV + AAV** | Kofoed | Mol Ther Methods Clin Dev | 2022 | 36284767 |
| **FUS + IV + AAV** | Owusu-Yaw | Pharmaceutics | 2024 | 38931834 |
| **FUS + IV + AAV** | Hsu | Plos One | 2013 | 23460893 |
| **FUS + IV + AAV** | Weber-Adrian | Sci. Rep | 2021 | 33479314 |
| **FUS + IV + AAV** | Blesa | Sci Adv. | 2023 | 37075119 |
| **FUS + IV + AAV** | Kofoed | Mol Ther Methods Clin Dev | 2021 | 34761053 |
| **FUS + IV + AAV** | Todd | ASGCT Proceedings | 2024 | -- |
| **FUS + IV + AAV** | Todd | ASGCT Proceedings | 2025 | -- |


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## Declaration of Interests

Dr. Bei, Dr. Todd and Dr. McDannold filed a provisional patent for FUS + AAV.CPP.16 and FUS + AAV.CPP.21 on 06/02/2023: “Combinatorial Use of Adeno-Associated Viruses and Focused Ultrasound for Overcoming the Blood-Brain-Barrier”; Provisional filed and patent pending.

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